



Water body extraction from satellite data

The architecture and the way it works

The architecture of LMMS is designed to adhere to the standards specified by the Open Geospatial Consortium's (OGC) Open Web-service Standard (OWS). As per OWS, once a platform-independent data model has been developed, the designs of the information model, the processing model and user interface model have to be taken up. These feed in modelling management services, processing services and human interaction services, respectively. This helps in building and deploying service interfaces for all the workflows of the user groups.

Landsat satellite data owned by NASA is in the open domain. Any registered user of the Earthexplorer website can download this data, which has two shortwave infrared bands along with one near infrared band. Water absorbs radiation and so the infrared spectrum is useful in distinguishing water from land. But in the shadow regions of forest areas there could be higher transpiration and thus there may also be heat-sinks that mimic water behaviour. To avoid this behaviour in computing NDWI (normalised difference water index) the team uses an infrared component, which is a composition of all the three infrared bands of landsat data.

After downloading the landsat images, the optimised NDWI is calculated using the following formula.

A vector output is generated after segmentation and thresholding the output of a rule based classification algorithm. This output is generated from time to time to understand the changes that happen in both space and time. Water tanks are ranked based on these changes and rejuvenation activities are planned based on the priorities.

After the data on waterbodies is extracted, it is stored in a central database by giving a unique ID generated based on the location of these bodies. Before going to the field, engineers/citizens/conservationists can navigate online to the area they are planning to visit using the interface provided. The water tanks in that area are shown in their mobile phones. Details of rejuvenation activities taken up on the tanks can be captured through the app and uploaded to the central database. Spatial temporal correlation can be done on the data sets to assess the impact of the rejuvenation activities taken up. This framework thus helps to identify water bodies, assess their current state, take up rejuvenation, and monitor the progress of these activities with the help of both

satellite data and field data.

Challenges

Gollapalli says that developing this system came with its share of challenges. The first challenge was to apply algorithms and build a case study to show the changes in water bodies and the stress on them. Getting funding to develop this kind of a system was also a challenge. After receiving government funding to develop this platform, the team is looking at a crowdsourced model to sustain it.

He adds, "Another issue was to make stakeholders understand that spatial data will be useful for planning statewide conservation activities. We worked with citizen groups to increase awareness, and by creating case studies we convinced the government to take it up on a large scale. Now we are trying to create a platform where people can learn and contribute. Here our challenge is to reach out to people. Before the onset of the pandemic we were conducting physical workshops with teams of researchers and conservationists, but now we are trying to build virtual platforms where people can participate in generating and using data. We are looking for investments in developing this platform."

$$\text{OptimizedNDWI} = f \left(B3, p \left(\sum_{i=5}^7 B^i \right) \right) \text{ where } f(i, j) = (i + j) / (i - j); p(x) = X * \begin{bmatrix} K11 & K12 \\ K21 & K22 \end{bmatrix}$$